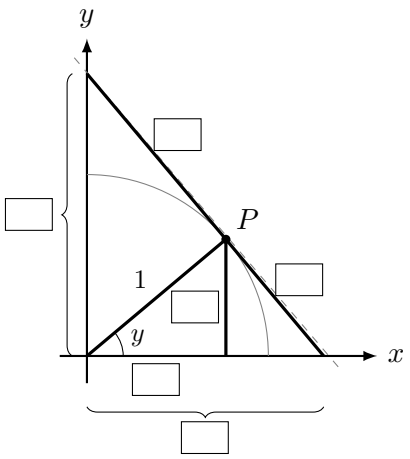
§3.5 Entry Ticket: Inverse Trig

Name: _____ Date: _____

1. The fundamental trignograph. The line is tangent to the unit circle at P. Write cos, sin, tan, cot, sec and csc in the boxes.



2. Use the Pythagorean theorem on Triangle II (sides 1, $\tan y$, $\sec y$) to write an identity.

3. Let y be the inside function. Draw the triangle, then simplify.

A. $\cos\left(\sin^{-1}\left(\frac{5}{6}\right)\right)$ **B.** $\cot\left(\csc^{-1}\left(\frac{4}{3}\right)\right)$

4. Write $\csc\left(\cos^{-1}(x)\right)$ as an algebraic expression in x.

5. Find each value, or write "does not exist": $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$ $\cos^{-1}\left(-\frac{1}{2}\right)$ $\sin^{-1}(2)$